전산화학 (2011년 1학기)

2차 시험 해답

(**총 8 문제, 74점**)

1. **(총 6 점, 각 2 점)**

   (a) $y' = -2xe^{-x^2}(x^2+1) + 2xe^{-x^2} = -2x^3e^{-x^2}$

   $y'' = -6x^2e^{-x^2} + 4x^4e^{-x^2}$

   (b) $y' = -e^{-x}\cos(e^{-x})$

   $y'' = e^{-x}\cos(e^{-x}) - e^{-2x}\sin(e^{-x})$

   (c) $y' = 2x\ln x + x$

   $y'' = 2\ln x + 3$

2. **(총 12점, 각 6점)**

   (a)

$$
\begin{aligned}
f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= 2xe^{-y^2} \\
f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= -2x^2ye^{-y^2} \\
f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y &= 2e^{-y^2} \\
f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x &= -2x^2ye^{-y^2} + 4x^2y^2e^{-y^2} \\
f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y \partial x}\right) &= -4xye^{-y^2} \\
f_{yx}(x,y) = \left(\frac{\partial^2 f}{\partial x \partial y}\right) &= -4xye^{-y^2}
\end{aligned}
$$

   (b)

$$
\begin{aligned}
f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= -ye^{-y}\sin(xy) \\
f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= -e^{-y}\cos(xy) - xe^{-y}\sin(xy) \\
f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y &= -y^2e^{-y}\cos(xy) \\
f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x &= e^{-y}\cos(xy) + 2xe^{-y}\sin(xy) - x^2e^{-y}\cos(xy) \\
f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y \partial x}\right) &= -e^{-y}\sin(xy) + ye^{-y}\sin(xy) - xye^{-y}\cos(xy) \\
f_{yx}(x,y) = \left(\frac{\partial^2 f}{\partial x \partial y}\right) &= -e^{-y}\sin(xy) + ye^{-y}\sin(xy) - xye^{-y}\cos(xy)
\end{aligned}
$$

3. **(총 12 점, 각 4점)**

(a) $df(x,y) = (x+y)^{-1}dx + (x+y)^{-1}dy$

$$M(x,y) = \frac{1}{x+y}, \qquad N(x,y) = \frac{1}{x+y}$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = -\frac{1}{(x+y)^2}$$
$$N_x(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = -\frac{1}{(x+y)^2}$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = \ln(x+y) + C$

(b) $df(x,y) = ay\sin(xy)dx + ax\sin(xy)dy \Rightarrow$ 완전미분 $f(x,y) = -a\cos(xy) + C$

$$M(x,y) = ay\sin(xy), \qquad N(x,y) = ax\sin(xy)$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = a\sin(xy) + axy\cos(xy)$$
$$N_y(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = a\sin(xy) + axy\cos(xy)$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = -a\cos(xy) + C$

(c) $df(T,V) = C_V(T)dT + \frac{RT}{V}dV$

$$M(T,V) = C_V(T), \qquad N(x,y) = \frac{RT}{V}$$
$$M_V(T,V) = \left(\frac{\partial M}{\partial V}\right)_T = 0$$
$$N_T(T,V) = \left(\frac{\partial N}{\partial T}\right)_V = \frac{R}{V}$$
$$M_V(T,V) \neq N_T(T,V)$$

불완전 미분

4. **(10점)**

$$c(x,t) = \frac{1}{\sqrt{(4\pi Dt)}}\exp\left[-\frac{x^2}{4Dt}\right]$$

이므로

$$\begin{aligned}\frac{\partial c}{\partial t} &= -\frac{1}{2\sqrt{4\pi Dt^3}}\exp\left[-\frac{x^2}{4Dt}\right] + \frac{1}{\sqrt{(4\pi Dt)}}\exp\left[-\frac{x^2}{4Dt}\right]\frac{x^2}{4Dt^2} \\ &= \left(\frac{x^2}{4Dt} - \frac{1}{2}\right)\frac{1}{\sqrt{4\pi Dt^3}}\exp\left[-\frac{x^2}{4Dt}\right] \qquad (1)\end{aligned}$$

이고,

$$\begin{aligned}
\frac{\partial c}{\partial x} &= -\frac{x}{2Dt}\frac{1}{\sqrt{(4\pi Dt)}}\exp\left[-\frac{x^2}{4Dt}\right] \\
\frac{\partial^2 c}{\partial x^2} &= -\frac{1}{2Dt}\frac{1}{\sqrt{(4\pi Dt)}}\exp\left[-\frac{x^2}{4Dt}\right] + \frac{x^2}{4D^2t^2}\frac{1}{\sqrt{(4\pi Dt)}}\exp\left[-\frac{x^2}{4Dt}\right] \\
&= \left(\frac{x^2}{4Dt}-\frac{1}{2}\right)\frac{1}{D\sqrt{4\pi Dt^3}}\exp\left[-\frac{x^2}{4Dt}\right] \qquad (2)
\end{aligned}$$

그러므로 (1) 식과 (2) 식으로부터

$$\begin{aligned}
\frac{\partial c}{\partial t} - D\left(\frac{\partial^2 c}{\partial x^2} - S\right) &= \left(\frac{x^2}{4Dt}-\frac{1}{2}\right)\frac{1}{\sqrt{4\pi Dt^3}}\exp\left[-\frac{x^2}{4Dt}\right] \\
&\quad -D\left\{\left(\frac{x^2}{4Dt}-\frac{1}{2}\right)\frac{1}{D\sqrt{4\pi Dt^3}}\exp\left[-\frac{x^2}{4Dt}\right] - S\right\} \\
&= \left(\frac{x^2}{4Dt}-\frac{1}{2}\right)\frac{1}{\sqrt{4\pi Dt^3}}\exp\left[-\frac{x^2}{4Dt}\right] \\
&\quad -\left(\frac{x^2}{4Dt}-\frac{1}{2}\right)\frac{1}{\sqrt{4\pi Dt^3}}\exp\left[-\frac{x^2}{4Dt}\right] + DS \\
&= DS
\end{aligned}$$

5. **(총 10점, 각 5점)**

(a) 문제에서 다음과 같이 주어졌으므로

$$\begin{aligned}
X &= \frac{m_a x_a + m_2 x_b}{m_a + m_2}, \qquad (3) \\
x &= x_a - x_b, \qquad (4)
\end{aligned}$$

다음과 같은 일차 편도함수를 얻을 수 있다.

$$\begin{aligned}
\left(\frac{\partial X}{\partial x_a}\right)_{x_b} &= \frac{m_a}{m_a + m_b}, \\
\left(\frac{\partial x}{\partial x_a}\right)_{x_b} &= 1, \\
\left(\frac{\partial X}{\partial x_b}\right)_{x_a} &= \frac{m_b}{m_b + m_b}, \\
\left(\frac{\partial x}{\partial x_b}\right)_{x_a} &= -1.
\end{aligned}$$

연쇄법칙을 이용하면 다음과 같은 결과를 얻는다.

$$\begin{aligned}
\frac{\partial}{\partial x_a} &= \frac{\partial X}{\partial x_a}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_a}\frac{\partial}{\partial x}, \\
&= \frac{m_a}{m_a + m_b}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}, \qquad (5)
\end{aligned}$$

그리고,

$$\begin{aligned}\frac{\partial}{\partial x_b} &= \frac{\partial X}{\partial x_b}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_b}\frac{\partial}{\partial x}, \\ &= \frac{m_b}{m_b + m_b}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}. \end{aligned} \tag{6}$$

(b) (5) 식으로부터,

$$\begin{aligned}\frac{\partial^2}{\partial x_a^2} &= \frac{\partial}{\partial x_a}\cdot\frac{\partial}{\partial x_a}, \\ &= \frac{\partial}{\partial x_a}\left(\frac{m_a}{m_a + m_b}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}\right). \end{aligned} \tag{7}$$

g 를 다음과 같이 정의한다면,

$$g \equiv \frac{m_a}{m_a + m_b}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}, \tag{8}$$

(7) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}\frac{\partial^2}{\partial x_a^2} &= \frac{\partial}{\partial x_a}\cdot\frac{\partial}{\partial x_a}, \\ &= \frac{\partial}{\partial x_a}\left(\frac{m_a}{m_a + m_b}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}\right), \\ &= \frac{\partial g}{\partial x_a}, \\ &= \frac{\partial X}{\partial x_a}\frac{\partial g}{\partial X} + \frac{\partial x}{\partial x_a}\frac{\partial g}{\partial x}, \qquad \text{(from Eq. (5))} \\ &= \frac{m_a}{m_a + m_b}\frac{\partial g}{\partial X} + \frac{\partial g}{\partial x}, \qquad \text{(from Eqs. (5) and (5))} \\ &= \frac{m_a}{m_a + m_b}\frac{\partial}{\partial X}\left(\frac{m_a}{m_a + m_b}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}\right) + \frac{\partial}{\partial x}\left(\frac{m_a}{m_a + m_b}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}\right), \\ &= \boxed{\frac{m_a^2}{(m_a + m_b)^2}\frac{\partial^2}{\partial X^2} + \frac{2m_a}{m_a + m_b}\frac{\partial^2}{\partial X \partial x} + \frac{\partial^2}{\partial x^2}.} \end{aligned} \tag{9}$$

같은 방법으로 (6) 식으로부터,

$$\begin{aligned}\frac{\partial^2}{\partial x_b^2} &= \frac{\partial}{\partial x_b}\cdot\frac{\partial}{\partial x_b}, \\ &= \frac{\partial}{\partial x_b}\left(\frac{m_b}{m_a + m_b}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}\right). \end{aligned} \tag{10}$$

h 를 다음과 같이 정의한다면,

$$h \equiv \frac{m_b}{m_a + m_b}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}, \tag{11}$$

(10) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial x_b^2} &= \frac{\partial}{\partial x_b}\cdot\frac{\partial}{\partial x_b}, \\
&= \frac{\partial}{\partial x_b}\left(\frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right), \\
&= \frac{\partial h}{\partial x_b}, \\
&= \frac{\partial X}{\partial x_b}\frac{\partial h}{\partial X}+\frac{\partial x}{\partial x_b}\frac{\partial h}{\partial x}, \qquad \text{(from Eq. (6))} \\
&= \frac{m_b}{m_a+m_b}\frac{\partial h}{\partial X}+\frac{\partial h}{\partial x}, \qquad \text{(from Eqs. (5) and (5))} \\
&= \frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}\left(\frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right)+\frac{\partial}{\partial x}\left(\frac{m_b}{m_a+m_b}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right), \\
&= \boxed{\frac{m_b^2}{(m_a+m_b)^2}\frac{\partial^2}{\partial X^2}-\frac{2m_b}{m_a+m_b}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}.} \qquad (12)
\end{aligned}$$

6. **(총 10점, 각 5점)**

(a) $f(x,y)$의 임계점을 (x^*,y^*)라 하면, 임계점에서

$$\begin{aligned}
f_x(x^*,y^*) &= 0 \Rightarrow 6x^*+12=0 \Rightarrow x^*=-2 \\
f_y(x^*,y^*) &= 0 \Rightarrow 12(y^*)^2-12y^*=0 \Rightarrow y^*(y^*-1)=0 \Rightarrow y^*=0 \quad \text{또는} \quad y^*=1
\end{aligned}$$

위 두 결과로부터 임계점은 $(-2,0)$과 $(-2,1)$이다.

(b) Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*,y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*,y^*)f_{yy}(x^*,y^*)-f_{xy}(x^*,y^*)^2.$$

주어진 함수 $f(x,y)=3x^2+12x+4y^3-6y^2+5$에 대하여

$$\begin{aligned}
f_{xx}(x,y) &= 6 \\
f_{yy}(x,y) &= 24y-12 \\
f_{xy}(x,y) &= 0
\end{aligned}$$

i. $(-2,0)$에서

$$\begin{aligned}
f_{xx}(-2,0) &= 6 \\
f_{yy}(-2,0) &= -12 \\
f_{xy}(-2,0) &= 0
\end{aligned}$$

그러므로

$$\Delta_1 = 6^2 = 36 > 0 \quad \text{and} \quad \Delta_2 = (6)\times(-12)-(0)^2 = -72 < 0$$

$\Delta_2 < 0$ 이므로 $f(x,y)$ 는 $(-2,0)$ 에서 안장점을 갖는다.

ii. $(-2,1)$ 에서

$$\begin{aligned} f_{xx}(-2,1) &= 6 \\ f_{yy}(-2,1) &= 12 \\ f_{xy}(-2,1) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = 6^2 = 36 > 0 \quad \text{and} \quad \Delta_2 = (6)\times(12) - (0)^2 = 72 > 0$$

$\Delta_1 > 0$ 이고 $\Delta_2 > 0$ 이므로 $f(x,y)$ 는 $(-2,1)$ 에서 극소값을 갖는다.

7. **(총 8점, 각 4점)**

(a)

$$\lim_{x\to 0} \frac{e^{-x}\sin^2 x}{x^2}$$

$x \to 0$ 일 때, 위 극한의 분자와 분모 모두 0에 수렴하므로 l'Hôpital의 법칙을 이용할 수 있다.

$$\lim_{x\to 0} \frac{e^{-x}\sin^2 x}{x^2} = \lim_{x\to 0} \frac{-e^{-x}\sin^2 x + e^{-x}2\sin x\cos x}{2x} = \lim_{x\to 0} \frac{(-\sin^2 x + e^{-x}\sin(2x))e^{-x}}{2x}$$

$x \to 0$ 일 때, 위 극한의 분자와 분모 모두 0에 수렴하므로 다시 l'Hôpital의 법칙을 이용하면

$$\begin{aligned} \lim_{x\to 0} \frac{e^{-x}\sin^2 x}{x^2} &= \lim_{x\to 0} \frac{(-\sin^2 x + \sin(2x))e^{-x}}{2x} \\ &= \lim_{x\to 0} \frac{(-2\sin x\cos x + 2\cos(2x))e^{-x} + (-\sin^2 x + \sin(2x))(-e^{-x})}{2} \\ &= 1 \end{aligned}$$

(b)

$$\lim_{x\to 0} \frac{\ln(1+x) - x}{x^2}$$

$x \to 0$ 일 때, 위 극한의 분자와 분모 모두 0에 수렴하므로 l'Hôpital의 법칙을 이용할 수 있다.

$$\begin{aligned} \lim_{x\to 0} \frac{\ln(1+x) - x}{x^2} &= \lim_{x\to 0} \frac{1/(1+x) - 1}{2x} && (13) \\ &= \lim_{x\to 0} -\frac{1}{2(1+x)} = -\frac{1}{2} && (14) \end{aligned}$$

또는, $x \to 0$ 일 때, (13) 식의 분자와 분모 모두 0에 수렴하므로 다시 l'Hôpital의

법칙을 이용하면

$$\begin{aligned}\lim_{x\to 0}\frac{\ln(1+x)-x}{x^2} &= \lim_{x\to 0}\frac{1/(1+x)-1}{2x}\\ &= \lim_{x\to 0}\frac{-1/(1+x)^2}{2}\\ &= -\frac{1}{2}\end{aligned}$$

8. **(6점)**

$$U(T) = \frac{N\epsilon e^{-\epsilon/(k_BT)}}{1+e^{-\epsilon/(k_BT)}}$$

이므로

$$\begin{aligned}C(T)=\frac{dU}{dT} &= \frac{d}{dT}\left(\frac{N\epsilon e^{-\epsilon/(k_BT)}}{1+e^{-\epsilon/(k_BT)}}\right)\\ &= N\epsilon\frac{e^{-\epsilon/(k_BT)}(\frac{\epsilon}{k_BT^2})(1+e^{-\epsilon/(k_BT)})-e^{-\epsilon/(k_BT)}e^{-\epsilon/(k_BT)}(\frac{\epsilon}{k_BT^2})}{(1+e^{-\epsilon/(k_BT)})^2}\\ &= \frac{N\epsilon^2}{k_BT^2}\frac{e^{-\epsilon/(k_BT)}}{(1+e^{-\epsilon/(k_BT)})^2}\end{aligned}$$